The integer-L^1 endpoints in higher-order affine Sobolev inequalities

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Abstract

Bullion–Gauthier's Theorems 1.1 and 1.2 leave open the cases in which the upper smoothness is an integer $m \geq 2$ and the upper exponent is 1. We close both omissions. The affine energy is the reciprocal-volume scale of the star body

$$\mathcal{K}_f = \{v \in \mathbb{R}^N : \|\partial_v^m f\|_{L^1} \leq 1\}.$$

A maximal-determinant tuple in this body yields a unimodular basis whose total pure-derivative cost is $O(\mathscr{E}_{m,1}(f))$. In those coordinates the pure power operator is elliptic and canceling, so Van Schaftingen's limiting Sobolev estimates give exactly the missing lower norms. No mixed-derivative estimate is used, and Ornstein's non-inequality is bypassed.

1 Source statements and omitted cases

Let $N \geq 2$. For an integer $m \geq 1$, the source [1] defines

$$a_f(v) := \|\partial_v^m f\|_{L^1(\mathbb{R}^N)}, \qquad \mathscr{E}_{m,1}(f) = \sigma_N^{(N+m)/N} \left(\int_{\mathbb{S}^{N-1}} a_f(\theta)^{-N/m} \, d\mathcal{H}^{N-1}(\theta) \right)^{-m/N},$$

where $\sigma_N = \mathcal{H}^{N-1}(\mathbb{S}^{N-1})$.

Theorem 1.1 of [1] states the strong critical affine Sobolev inequality possibly except when $s \geq 2$ is an integer and $p = 1$. The definition and exception occur on source PDF page 2.

Theorem 1.2 compares two affine energies on the Sobolev line $s_2 - N/p_2 = s_1 - N/p_1$, again possibly except when the upper pair is an integer $s_2 \geq 2$ and $p_2 = 1$. This appears on source PDF page 3.

Thus the two extracted questions are whether, for every integer $m \geq 2$,

$$\|f\|_{L^{N/(N-m)}} \lesssim \mathscr{E}_{m,1}(f) \quad (m < N),$$

and, more generally, whether

$$\mathscr{E}_{s,p}(f) \lesssim \mathscr{E}_{m,1}(f) \quad \text{when} \quad 0 < s < m, \qquad m - N = s - \frac{N}{p}.$$

2 Proof intuition

The source route tries to control a full order-m Sobolev tensor after an optimal unimodular coordinate change. At $p = 1$, mixed derivatives make this incompatible with Ornstein's non-inequality.

where the best constant $C_{s,p,N}$ is given by an explicit formula involving a best Sobolev constant $\widetilde{C}_{s,p,N}$ (similarly to above). (Here, $\Delta_h f(x) := f(x+h) - f(x)$.) Their sharp result implies, by extrapolation ($s \to 1^-$), (1.2) and its extension to $\mathring{W}^{1,p}$. In a related direction, a new approach to affine Moser-Trudinger inequalities was proposed in [6].

We now present our contributions. *The main goal of this article is to obtain affine Sobolev inequalities of general smoothness order s (not necessarily ≤ 1).* Since, when $N = 1$, affine Sobolev inequalities coincide with standard Sobolev inequalities, *in what follows we always assume that $N \geq 2$, unless otherwise stated.*

Given $f \in \dot{W}^{s,p}(\mathbb{R}^N)$ (for the definition of $\dot{W}^{s,p}(\mathbb{R}^N)$, see (2.3) and (2.4)), we denote

$$\mathscr{E}_{s,p}(f) := \sigma_N^{(N+sp)/Np} \left(\int_{\mathbb{S}^{N-1}} \left(\int_0^\infty t^{-sp-1} \left\| \Delta_{t\xi}^{\lfloor s \rfloor + 1} f \right\|_{L^p(\mathbb{R}^N)}^p dt \right)^{-N/sp} d\mathscr{H}^{N-1}(\xi) \right)^{-s/N},$$

if $s > 0$ is not an integer, respectively

$$\mathscr{E}_{s,p}(f) := \sigma_N^{(N+sp)/Np} \left(\int_{\mathbb{S}^{N-1}} \left(\int_{\mathbb{R}^N} \left| \partial_\xi^s f(x) \right|^p dx \right)^{-N/sp} d\mathscr{H}^{N-1}(\xi) \right)^{-s/N},$$

if $s \geq 1$ is an integer. Here, σ_N is the surface area of the unit sphere $\mathbb{S}^{N-1}$.

Our first main results are the following.

Theorem 1.1. Let $s > 0$ and $1 \leq p < \infty$ satisfy $sp < N$. Then there exists $K = K_{s,p,N} < \infty$ such that

$$\|f\|_{L^{Np/(N-sp)}(\mathbb{R}^N)} \leq K \mathscr{E}_{s,p}(f), \ \forall f \in \mathring{W}^{s,p}(\mathbb{R}^N), \tag{1.4}$$

possibly except when $s \geq 2$ is an integer and $p = 1$.

Figure 1: The integer affine energy and the omitted case in Theorem 1.1 of [1], source PDF page 2.

The desired conclusions need only N pure derivatives. For a fixed basis these form an elliptic, canceling system, and limiting Sobolev theory controls the critical lower norm directly.

It remains to choose the basis. The function $a_f(v) = \|\partial_v^m f\|_1$ is m-homogeneous, so its unit sublevel set is a symmetric star body. Its radial volume is exactly the negative spherical mean defining $\mathscr{E}_{m,1}$. A maximal-determinant argument, requiring no convexity, extracts a unimodular basis with total cost $O(\mathscr{E}_{m,1})$. The geometry and the canceling estimate have precisely the same critical scaling.

3 The two ingredients

Lemma 1 (Maximal-determinant selection). *Suppose a_f is continuous and positive on $\mathbb{S}^{N-1}$. There is a basis $v_1, \ldots, v_N$ with $\det(v_1, \ldots, v_N) = 1$ and*

$$\sum_{i=1}^N a_f(v_i) \leq C_{N,m} \mathscr{E}_{m,1}(f).$$

Theorem 1.2. Let $0 < s_1 < s_2$ and $1 \leq p_1, p_2 < \infty$ satisfy

$$s_2 - \frac{N}{p_2} = s_1 - \frac{N}{p_1}. \tag{1.5}$$

Then there exists $K = K_{s_1,p_1,s_2,p_2,N} < \infty$ such that

$$\mathscr{E}_{s_1,p_1}(f) \leq K \mathscr{E}_{s_2,p_2}(f),\ \forall\, f \in \dot{W}^{s_1,p_1}(\mathbb{R}^N) \cap \dot{W}^{s_2,p_2}(\mathbb{R}^N), \tag{1.6}$$

possibly except when $s_2 \geq 2$ is an integer and $p_2 = 1$.

We emphasize the fact that our approach is new, even in the known case where $0 < s \leq 1$. One of its features is that, while it encompasses the case $0 < s \leq 1$, it does not provide the sharp constants in (1.3). The trade-off is that we gain in generality, but lose in precision. This

Figure 2: The omitted case in Theorem 1.2 of [1], source PDF page 3.

Proof. Set $\mathcal{K}_f = \{v : a_f(v) \leq 1\}$. It is a compact symmetric star body with nonempty interior, and polar coordinates give

$$|\mathcal{K}_f| = \frac{1}{N} \int_{\mathbb{S}^{N-1}} a_f(\theta)^{-N/m}\, d\mathcal{H}^{N-1}(\theta), \qquad |\mathcal{K}_f|^{-m/N} = N^{m/N} \sigma_N^{-(N+m)/N} \mathscr{E}_{m,1}(f). \tag{1}$$

Choose $w_1, \ldots, w_N \in \mathcal{K}_f$ maximizing $d = |\det(w_1, \ldots, w_N)| > 0$. If $x = \sum_i \alpha_i w_i \in \mathcal{K}_f$, replacing column i by x gives $|\alpha_i|\, d \leq d$; hence $|\alpha_i| \leq 1$. Therefore

$$\mathcal{K}_f \subseteq \left\{ \sum_i \alpha_i w_i : |\alpha_i| \leq 1 \right\}, \qquad |\mathcal{K}_f| \leq 2^N d.$$

After changing one sign, set $v_i = d^{-1/N} w_i$. The determinant is one and

$$\sum_i a_f(v_i) = d^{-m/N} \sum_i a_f(w_i) \leq N d^{-m/N} \leq N 2^m \left|\mathcal{K}_f\right|^{-m/N}.$$

Now use (1). $\square$

Lemma 2 (Pure powers form a canceling system)**.** *For $g \in C_c^\infty(\mathbb{R}^N)$:*

1. *if $m < N$, then*
$$\|g\|_{L^{N/(N-m)}} \leq C_{N,m} \sum_{i=1}^N \|\partial_i^m g\|_{L^1};$$
2. *if $0 < s < m$, $1 < p < \infty$, and $1/p - s/N = 1 - m/N$, then*
$$|g|_{\dot{W}^{s,p}} \leq C_{N,m,s,p} \sum_{i=1}^N \|\partial_i^m g\|_{L^1},$$
with the integer Sobolev seminorm when s is integral and the Sobolev–Slobodeckij seminorm of [1] otherwise.

Proof. For $A(D)g = (\partial_1^m g, \dots, \partial_N^m g)$, the scalar symbol is

$$A(\xi)t = t(\xi_1^m, \dots, \xi_N^m).$$

It is elliptic. It is also canceling in the sense of [2], since

$$\bigcap_{\xi \neq 0} A(\xi)[\mathbb{R}] \subseteq \bigcap_{j=1}^{N} A(e_j)[\mathbb{R}] = \bigcap_{j=1}^{N} \operatorname{span}\{e_j\} = \{0\},$$

using $N \geq 2$. Van Schaftingen's Theorem 1.3 [2] gives

$$\left\|D^{m-1} g\right\|_{L^{N/(N-1)}} \leq C \sum_i \|\partial_i^m g\|_{L^1}.$$

The classical homogeneous Sobolev embedding of order $m-1$ gives part 1. For part 2, use Proposition 8.12 of [2] with $\dot{F}^s_{p,2} = \dot{W}^{s,p}$ when s is integral, and Proposition 8.13 with $\dot{B}^s_{p,p} = \dot{W}^{s,p}$ when s is nonintegral. Their parameter identity is exactly $1/p - s/N = 1 - m/N$. $\square$

4 Full endpoint theorem

Theorem 1. *Let $N \geq 2$ and let $m \geq 2$ be an integer.*

1. *If $m < N$, then*

$$\|f\|_{L^{N/(N-m)}(\mathbb{R}^N)} \leq C_{N,m} \mathscr{E}_{m,1}(f), \qquad f \in \mathring{W}^{m,1}(\mathbb{R}^N).$$

2. *If $0 < s < m$, $1 \leq p < \infty$, and $m - N = s - N/p$, then*

$$\mathscr{E}_{s,p}(f) \leq C_{N,m,s,p} \mathscr{E}_{m,1}(f), \qquad f \in \dot{W}^{s,p}(\mathbb{R}^N) \cap \dot{W}^{m,1}(\mathbb{R}^N).$$

Consequently the omitted integer-L^1 cases in both Theorems 1.1 and 1.2 of [1] hold.

Proof. First let $f \in C_c^\infty(\mathbb{R}^N)$, $f \neq 0$. Then a_f is continuous and positive on the sphere: if $a_f(v) = 0$, the restriction of f to almost every line parallel to v is a polynomial of degree at most $m-1$, and compact support forces it to vanish. Lemma 1 supplies $v_1, \dots, v_N$ such that

$$\det(v_1, \dots, v_N) = 1, \qquad \sum_i \left\|\partial_{v_i}^m f\right\|_{L^1} \leq C_{N,m} \mathscr{E}_{m,1}(f). \tag{2}$$

Let $T \in \mathrm{SL}_N$ have columns v_i, and set $g = f \circ T$. Then

$$\|\partial_i^m g\|_{L^1} = \left\|\partial_{v_i}^m f\right\|_{L^1}.$$

If $m < N$, Lemma 2 and (2) yield $\|g\|_{L^{N/(N-m)}} \leq C\mathscr{E}_{m,1}(f)$. The Lebesgue norm is unchanged because $\det T = 1$, proving part 1.

For part 2, the parameter relation and $s < m$ imply $p > 1$ and $s \in (m - N, m)$. Lemma 2 gives

$$|g|_{\dot{W}^{s,p}} \leq C \sum_i \left\|\partial_{v_i}^m f\right\|_{L^1} \leq C \mathscr{E}_{m,1}(f).$$

The source's Jensen estimate gives $\mathscr{E}_{s,p}(g) \le C\,|g|_{\dot{W}^{s,p}}$, while its affine-invariance proposition gives $\mathscr{E}_{s,p}(g) = \mathscr{E}_{s,p}(f)$. Hence

$$\mathscr{E}_{s,p}(f) = \mathscr{E}_{s,p}(g) \le C\,|g|_{\dot{W}^{s,p}} \le C\mathscr{E}_{m,1}(f).$$

Crucially, the ordinary seminorm is used only for g; it is the affine energy, not that seminorm, which is transported back through T.

For the source spaces, Appendix A of [1] gives $f_j \in C_c^\infty$ with $|f_j - f|_{\dot{W}^{m,1}} \to 0$. The standard homogeneous embedding in Appendix B gives convergence in the lower space (and in $L^{N/(N-m)}$ for part 1). Moreover,

$$\sup_{\theta\in\mathbb{S}^{N-1}} \left|a_{f_j}(\theta) - a_f(\theta)\right| \le C_{N,m}\,|f_j - f|_{\dot{W}^{m,1}} \longrightarrow 0.$$

The no-constant-direction lemma of [1] shows that a_f is positive on the sphere whenever the relevant lower seminorm is nonzero: otherwise $\partial_v^m f = 0$ would make every m-th finite difference along v zero. Compactness gives a positive minimum, so $\mathscr{E}_{m,1}$ is continuous under the displayed uniform convergence. Passing to the limit proves the claims. If the lower seminorm in part 2 is zero, then $\mathscr{E}_{s,p}(f) = 0$ and the conclusion is immediate. $\square$

5 Verification, scope, and novelty

The determinant normalization is exact, and the selection lemma does not assume convexity. Cancellation is checked from the symbol rather than inferred from ellipticity. In the cross-order case $p > 1$ is automatic, so the cited Triebel–Lizorkin/Besov estimates cover every parameter. The only coordinate-dependent norm is evaluated after the chosen change of variables; the final left side is SL_N-invariant. Thus no uniform control of mixed derivatives is hidden in the argument.

This result closes Theorems 1.1 and 1.2 only. It does not claim the omitted integer-L^1 minimizing-orbit coercivity in Theorem 1.4 or the reverse comparisons involving the full order-m tensor. Ornstein's obstruction remains relevant there.

The four cheap run indexes were searched for arXiv:2506.10473, its exact title, the integer-$p = 1$ endpoint, pure directional derivatives, canceling operators, and the maximal-determinant star-body mechanism. No duplicate was found. Bounded external searches on 2026-08-11 used the exact exception phrases and the same combinations. They found the current published source, which still contains both exceptions, and the general canceling-operator literature represented by [2], but no paper closing these affine endpoints.

The recommended verdict is *candidate full solution, likely valid, pending expert review.* The highest-value checks are the identification of the source seminorms with the spaces in Propositions 8.12–8.13, the homogeneous density passage, and the no-constant-direction step for general representatives.

References

[1] Tristan Bullion–Gauthier, *Higher-order affine Sobolev inequalities*, Journal of Functional Analysis **290** (2026), no. 11, 111419; arXiv:2506.10473v2. https://arxiv.org/abs/2506.10473

[2] Jean Van Schaftingen, *Limiting Sobolev inequalities for vector fields and canceling linear differential operators*, Journal of the European Mathematical Society **15** (2013), 877–921; arXiv:1104.0192. https://arxiv.org/abs/1104.0192